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MODELING OF CONVENTIONAL TENSILE TESTING OF ZIRCONIUM 2.5%NIOBIUM

A progress report submitted in partial
fulfilment of the CP303 project

by

Kewal Singh

(2022MMB1383)

Under the Guidance of

DR ABHISHEK TIWARI

DEPARTMENT OF METALLURGICAL AND MATERIALS ENGINEERING

INDIAN INSTITUTE OF TECHNOLOGY ROPAR

May 2026

Executive Summary

Modeling of Conventional Tensile Testing of Zirconium–2.5% Niobium Alloy

Background:

Conventional tensile testing is widely used for evaluating the mechanical behavior of Zr–2.5%Nb alloy in nuclear applications because it provides standardized and reliable measurements of properties such as yield strength, ultimate tensile strength, and elongation. These larger specimen tests offer better reproducibility and are less sensitive to geometric and microstructural variations compared to miniature samples.

Problem Statement:

This research aims to develop, validate, and characterize a conventional tensile test methodology for Zr–2.5%Nb alloy, bridging the gap between small-specimen data and conventional tensile modeling for this technologically critical nuclear material.

Hypothesis:

By optimizing finite element modeling parameters, specimen geometry, and constitutive material behavior, tensile deformation and failure characteristics of Zr–2.5%Nb alloy can be accurately predicted for both miniature and conventional specimens.

Conceptual Approach:

- Design 3D CAD models of miniature and conventional tensile specimens.
- Import specimen geometry into finite element software (ABAQUS).
- Define material properties using experimentally obtained stress–strain data.
- Apply appropriate boundary conditions and displacement-controlled loading.
- Validate numerical results against experimental tensile data.

Preliminary Work:

Initial specimen designs are complete and fabrication trials using established best practices are ongoing. Early tensile tests with calibrated video extensometry are underway, with results benchmarked against standard data. Experimental load–displacement data have been processed into true stress–true strain and true stress–plastic strain forms for ABAQUS constitutive input. Preliminary ABAQUS simulations show yield strength and UTS predictions within 1% of experimental values.

Deliverables:

- Validated finite element model for tensile testing of Zr–2.5%Nb alloy.
- Simulated stress–strain curves for miniature and conventional specimens.
- Comparison database between experimental and numerical results.
- Necking and failure analysis using ABAQUS.

Impact:

Successful modeling of tensile testing for Zr–2.5%Nb alloy will improve the reliability of miniature specimen testing for nuclear materials, supporting safer reactor operation and advancing constitutive modeling frameworks for advanced structural alloys.

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1. Introduction

Zirconium and its alloys are widely used in nuclear power applications due to their exceptional combination of low neutron absorption cross-section, good mechanical strength, and excellent corrosion resistance in high-temperature water and steam environments. Among these alloys, Zirconium–2.5% Niobium (Zr–2.5%Nb) is one of the most established materials for pressure tubes in CANDU (Canada Deuterium Uranium) reactors and several research reactor systems. The addition of niobium significantly enhances the alloy's mechanical properties, microstructural stability, and resistance to irradiation-induced degradation. Because pressure tubes and nuclear components experience continuous mechanical loading, thermal cycling, and neutron irradiation, accurate evaluation of their mechanical properties is essential for safety and performance assessments.

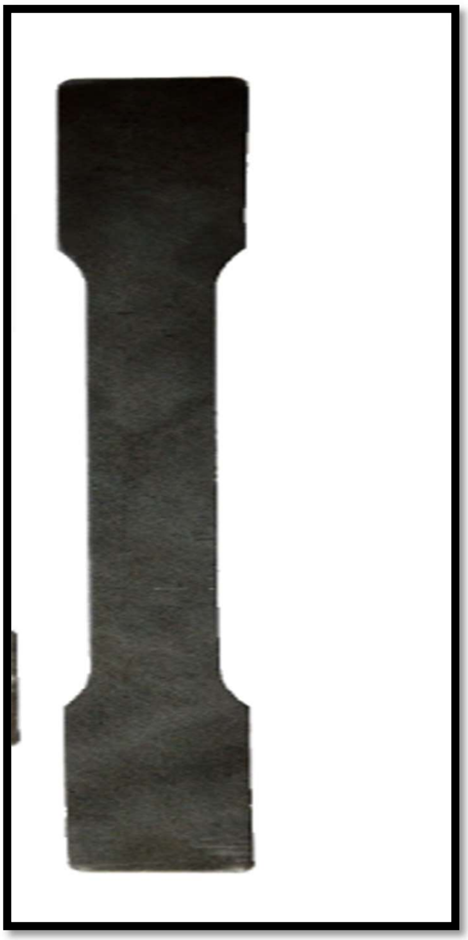

Fig 1 conventional tensile sample

Tensile testing is one of the most fundamental methods for determining the mechanical properties of Zr–2.5%Nb alloy, including yield strength, ultimate tensile strength (UTS), elastic modulus, and elongation. While conventional tensile testing provides standardized and reliable results, it requires relatively large specimen sizes and significant material volume. This becomes challenging when testing irradiated materials, localized regions, thin sections, or miniature components where material availability is limited.

To overcome these limitations, modeling and simulation of tensile testing have become increasingly important. Finite Element Modeling (FEM) enables detailed analysis of stress–strain behavior, strain localization, necking, and fracture mechanisms under tensile loading conditions. Modeling also helps establish correlations between miniature and conventional tensile specimens, reducing experimental cost and improving understanding of deformation behavior.

In the modeling of Zr–2.5%Nb alloy tensile testing, material constitutive behavior, specimen geometry, grain size effects, and boundary conditions play significant roles in accurately predicting mechanical response. Numerical simulations using software such as ABAQUS or ANSYS can reproduce experimental stress–strain curves and provide insight into localized deformation and failure mechanisms that are difficult to observe experimentally.

The present study focuses on developing and validating a reliable tensile test modeling methodology for Zr–2.5%Nb alloy. The work aims to compare miniature and conventional specimen behavior, evaluate the influence of geometry and microstructure, and improve the accuracy of mechanical property prediction for advanced nuclear material applications.

CHAPTER 2

Literature Review

2.1 Introduction to Zirconium Alloys for Nuclear Applications

Zirconium-based alloys have been central to nuclear reactor engineering for more than six decades due to their unique combination of low thermal neutron absorption cross-section, superior corrosion resistance, and acceptable mechanical properties at elevated temperatures. These characteristics make them ideal for components such as pressure tubes, calandria tubes, and fuel cladding. Among the various zirconium alloys, Zr–2.5%Nb (Zirconium–2.5 wt% Niobium) has gained exceptional importance, especially in CANDU reactors, where it is used extensively in pressure tubes operating under high pressures, elevated temperatures, and intense neutron flux.

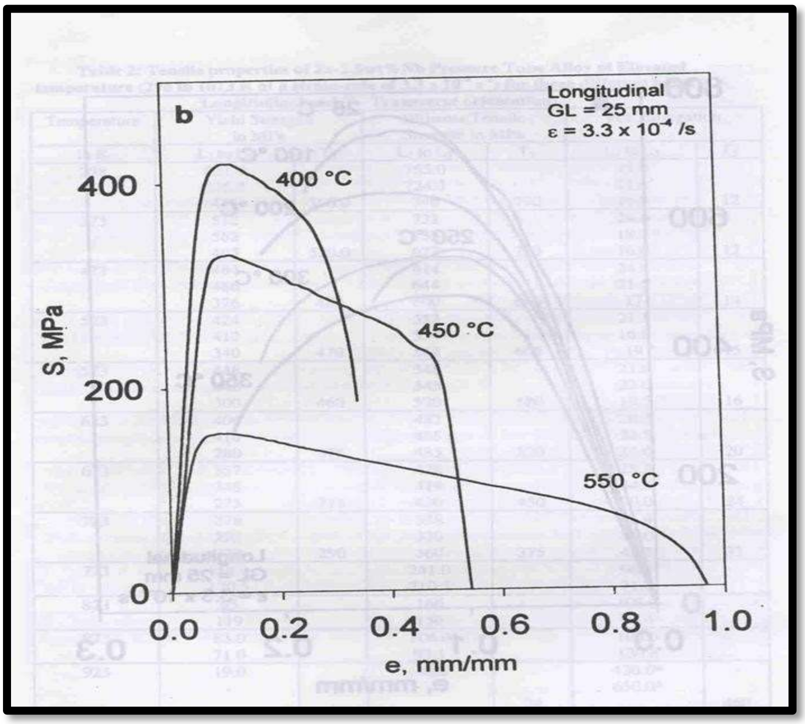

Fig2 stress vs strain curve under 400 to 500celsius temperature

Extensive research has been conducted on Zr–2.5%Nb to understand its microstructure, phase evolution, mechanical properties, irradiation behaviour, and in-service degradation mechanisms. However, as reactor components age, material sampling becomes more difficult. Researchers have therefore moved towards miniature mechanical testing, including miniature tensile testing, to characterize the properties of irradiated or service-degraded material with minimal material removal. This literature review compiles and analyzes previously published work on the alloy's metallurgy, testing challenges, miniature testing methodologies, the influence of irradiation, and recent technological advancements.

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2.2 Metallurgy and Microstructure of Zr–2.5%Nb Alloy

2.2.1 Alloy Composition and Phase Structure

Zr–2.5%Nb is a two-phase α + β alloy. The α-Zr phase (HCP) is the major phase responsible for good corrosion resistance, while the β-Nb phase (BCC) is the minor phase that improves strength and irradiation stability. Several researchers (Holt, 1970s; Coleman, 1980s) documented that niobium stabilizes the β phase, which remains as fine filaments or particles within the α matrix after thermomechanical processing. The α/β distribution depends strongly on cold work, heat treatment, neutron irradiation, and hydrogen concentr ation.

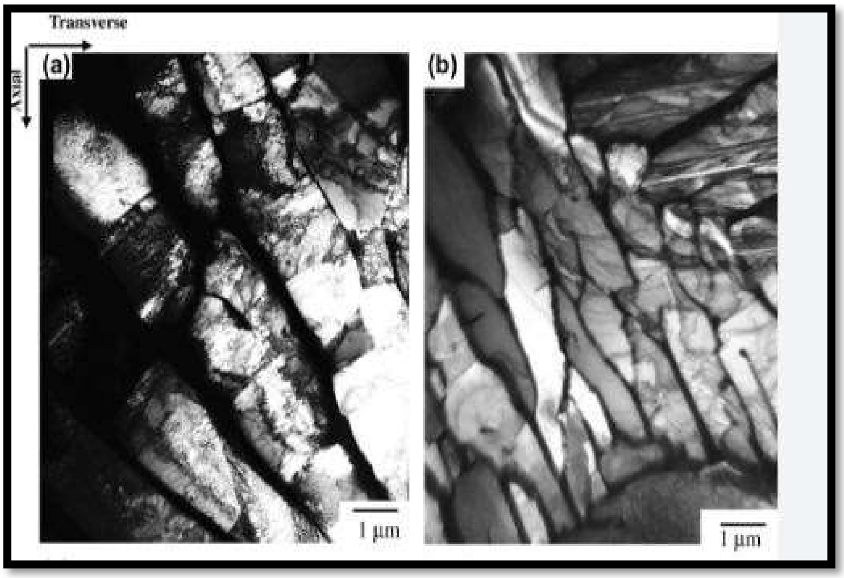

Fig3 microstructur image

2.3 Conventional Tensile Testing Methodology

3 6

Conventional tensile testing of Zr–2.5%Nb alloy is generally performed according to ASTM standards, with standardized specimen dimensions to ensure repeatability and reproducibility. Typical tensile specimen dimensions include a gauge length of 25 mm, width of 6.25 mm, and thickness of 3–4 mm. Tests are commonly performed using universal testing machines such as Instron machines equipped with resistance-heated furnaces, temperature control systems, extensometers, and video extensometry systems.

Researchers reported that specimens were soaked at the desired temperature before testing to ensure thermal equilibrium. Argon gas purging was used to minimize oxidation during elevated-temperature testing, and borax coatings were applied at temperatures above 550

°C to prevent excessive oxidation. The strain rate used in conventional tensile tests generally ranged between 10^{-5} /s and 10^{-3} /s. The tests provided stress–strain curves used to determine yield strength, ultimate tensile strength, total elongation, work hardening behavior, and fracture characteristics.

2.4 Tensile Behavior at Room Temperature

At room temperature, Zr–2.5%Nb alloy exhibits high strength and moderate ductility, with tensile deformation primarily controlled by dislocation slip and work hardening. Reported room-temperature properties include a yield strength of 550–650 MPa, UTS of 700–750 MPa, and elongation of 12–30%. The alloy exhibits strong work hardening during tensile deformation, resulting in a continuous increase in flow stress with strain. Transverse specimens generally exhibit higher strength compared to longitudinal specimens due to crystallographic texture effects.

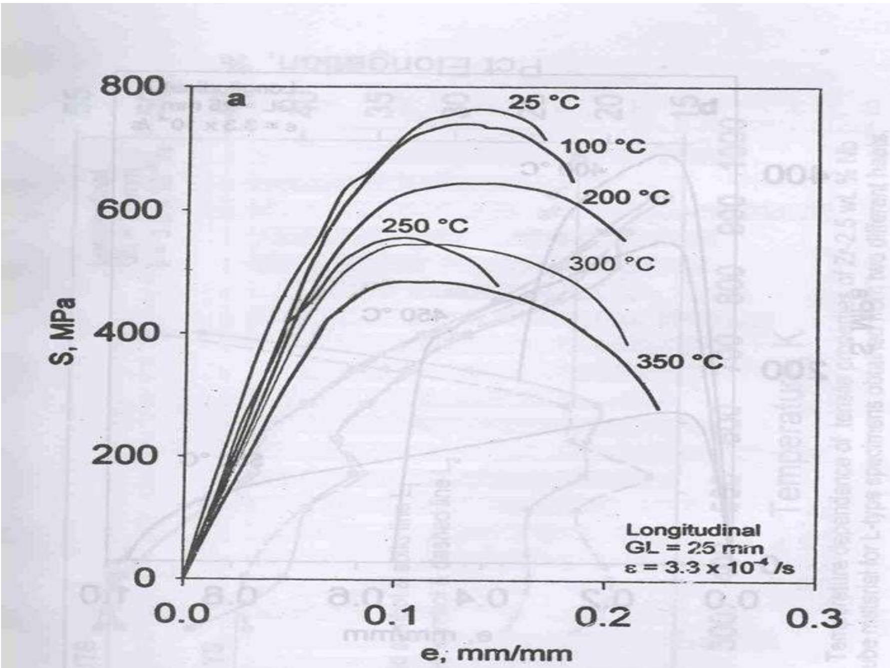

Fig4 different temperature

2.5 Effect of Temperature on Tensile Properties

Several researchers have extensively studied the effect of temperature on the tensile properties of Zr–2.5%Nb alloy in the range from room temperature to 800 °C. The key findings are: yield strength and UTS decrease continuously with increasing temperature; ductility increases significantly at elevated temperatures; deformation at temperatures below 350 °C is dominated by work hardening; and dynamic recovery and work softening become dominant above 350 °C. Negative slopes in stress–strain curves beyond the maximum stress point at elevated temperatures indicate dynamic recovery processes.

Near 800 °C, the yield strength is reduced to approximately 10 MPa, while elongation can exceed 400–700%. This increased ductility at high temperatures is attributed to enhanced atomic diffusion, dynamic recrystallization, and grain boundary sliding.

2.6 Hydride Formation and Embrittlement

35 Hydrogen absorption during service leads to hydride precipitation in Zr–2.5%Nb alloy. Hydride formation significantly affects tensile behavior and fracture properties by reducing ductility, fracture toughness, and crack resistance. Researchers reported that increasing hydrogen concentration significantly increases hydride volume fraction: 23 ppm hydrogen corresponds to 2.7% hydrides; 95 ppm to 7.8% hydrides; and 360 ppm to 30.3% hydrides. Hydrides act as brittle phases that may initiate crack nucleation, delayed hydride cracking, and brittle fracture. Hydride orientation also strongly influences tensile behavior.

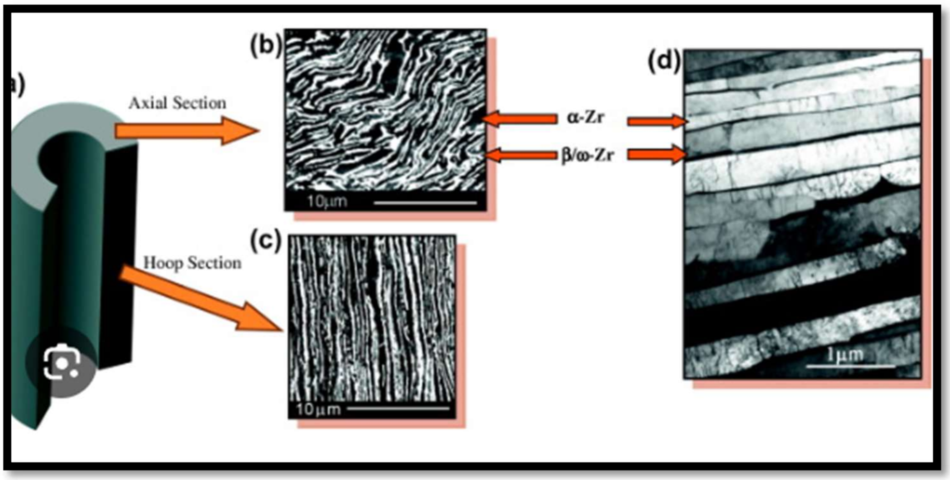

Fig5 hydride concentration

2.7 Anisotropy in Conventional Tensile Testing

38 Anisotropy is one of the most important characteristics of Zr–2.5%Nb alloy. Due to thermomechanical processing, the alloy develops strong crystallographic texture during fabrication. Researchers performed tensile tests on longitudinal, transverse, and radial specimens. Results showed that transverse specimens exhibited the highest strength while radial specimens showed the lowest strength but highest ductility. The anisotropic behavior was significant up to approximately 350 °C; beyond this temperature, differences between longitudinal and transverse properties became much smaller because dynamic recovery reduced texture influence. Accurate prediction of directional mechanical properties is therefore necessary for structural integrity assessment.

2.8 Dynamic Strain Aging

Researchers identified dynamic strain aging (DSA) in Zr–2.5%Nb alloy in the temperature range 200–300 °C. DSA was characterized by a plateau in yield strength, serrated flow behavior, and reduction in ductility. It occurs due to interaction between solute atoms and mobile dislocations. This phenomenon is especially important because reactor operating temperatures lie within this range, and DSA may therefore influence fatigue performance, fracture resistance, and crack initiation behavior.

2.9 Fracture Behavior

Conventional tensile testing revealed different fracture mechanisms at different temperatures. At lower temperatures, fracture was mainly ductile with microvoid coalescence as the dominant mechanism. At elevated temperatures, extensive necking occurred and dynamic recovery influenced the fracture mode. Hydride-containing specimens often showed brittle fracture features due to hydride cracking. Researchers used fractography and scanning electron microscopy (SEM) to investigate these failure mechanisms.

2.10 Finite Element Modeling of Conventional Tensile Tests

Finite Element Modeling (FEM) has been widely used to simulate conventional tensile testing of Zr–2.5%Nb alloy. Researchers used FEM to predict stress–strain response, analyze necking behavior, evaluate strain localization, study hydride effects, and predict fracture initiation. Dundulis et al. developed FEM models containing a zirconium matrix with hydride inclusions; the numerical results showed excellent agreement with experimental tensile data. The FEM studies improved understanding of localized stress distribution, hydride-induced embrittlement, and plastic strain evolution.

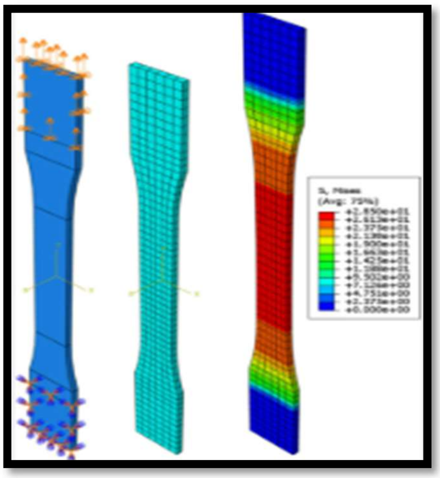

Fig6 modeling image

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2.11 Constitutive Modeling

Constitutive models are essential for accurate tensile simulations. Researchers converted engineering stress–strain data into true stress–strain form using the standard relations:

True stress: $\sigma t = \sigma e(1 + \varepsilon e)$

True strain: $\varepsilon t = \ln(1 + \varepsilon e)$

These equations improve simulation accuracy during plastic deformation. Because Zr–2.5%Nb alloy is anisotropic, researchers also used Hill's anisotropic yield criterion. Hill's criterion accurately represented directional yielding behavior and was applied within ABAQUS for multiaxial constitutive modeling. At elevated temperatures between 650–800 °C, the alloy exhibits extensive superplasticity, with reported elongation values of up to 650% for longitudinal specimens and 900% for transverse specimens; the strain-rate sensitivity coefficient (m-value) exceeded 0.5 in the superplastic region.

2.12 Factors Affecting Accuracy of Conventional Tensile Tests

Several factors influence the accuracy of conventional tensile testing results for Zr–2.5%Nb alloy:

- Machine compliance: Elastic deformation of the test machine produces errors in displacement measurement, requiring compliance correction, stiff frame design, and use of external extensometers or digital image correlation (DIC).

Fig7 break sample

- Heat treatment: Changes in grain structure, residual stress, and phase distribution affect tensile response.

- Surface condition: Rough surfaces may introduce stress concentration, cause premature crack initiation, and reduce elongation; polished specimens improve accuracy.
- Oxidation during testing: At elevated temperatures, oxidation reduces cross-sectional area and introduces surface cracks; argon atmosphere and borax coatings minimize this effect.
- Residual stress: Stresses produced during machining or fabrication may cause distortion, nonuniform yielding, and early failure; stress-relief heat treatment helps reduce residual stress.

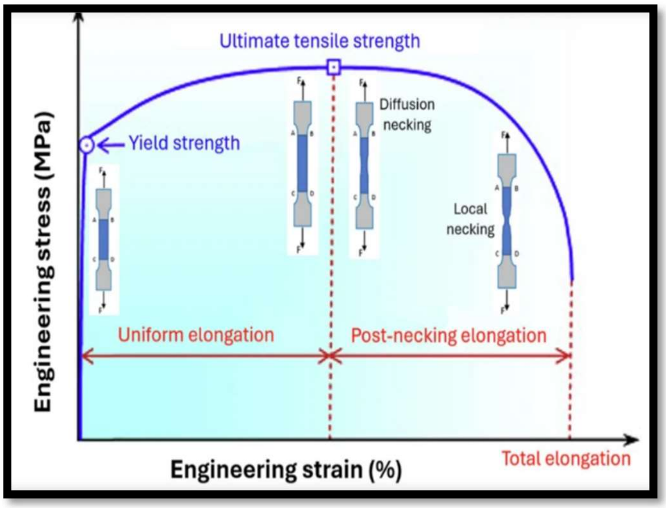

Fig8 how break

- Specimen geometry: Gauge length, width, thickness, and fillet radius all influence stress distribution, necking behavior, and uniform elongation; ASTM standard geometry improves repeatability.
- Irradiation effects: Neutron irradiation causes radiation hardening, reduced ductility, and defect accumulation in service.

2.13 Limitations of Conventional Tensile Testing

Although conventional tensile testing provides reliable results, several limitations exist: large material requirement; difficulty in testing irradiated materials; inability to evaluate localized regions; high machining cost; and difficulty in testing thin sections. These limitations motivated the development of miniature tensile testing methods. However, conventional tensile testing remains the benchmark for validation of miniature tensile tests, constitutive modeling, finite element analysis, and reactor safety evaluation.

Chapter 3

Preliminary Work

The requirement for accurate mechanical characterization of Zr–2.5%Nb alloy used in nuclear reactor pressure tubes has led to the integration of experimental tensile testing with finite element modeling. Conventional tensile testing provides reliable mechanical properties such as yield strength, UTS, ductility, and strain hardening behavior, while numerical modeling enables detailed understanding of stress distribution, strain localization, necking, and failure mechanisms.

In the present preliminary work, experimentally obtained tensile data from conventional tensile testing of Zr–2.5%Nb alloy were processed and converted into engineering stress–strain and true stress–strain relationships for constitutive modeling. The processed data were then prepared for finite element simulation of tensile deformation behavior.

3.1 Objectives of Preliminary Work

- Process experimental tensile data obtained from conventional tensile testing.
- Convert engineering stress–strain data into true stress–true strain data.
- Determine plastic strain for material constitutive modeling.
- Extract important mechanical properties: yield strength, UTS, elongation, and reduction in area.
- Prepare material input data for ABAQUS finite element simulation.
- Establish a validated methodology for future tensile modeling of Zr–2.5%Nb alloy.

3.2 Specimen Details

The conventional tensile specimen used in this study followed standard ASTM tensile geometry. Average specimen dimensions are given in Table 1.

Table 1: Dimensions for conventional tensile testing

Parameter	Value
Gauge Length (GL)	26.37 mm
Thickness	4.57 mm
Width	6.04 mm

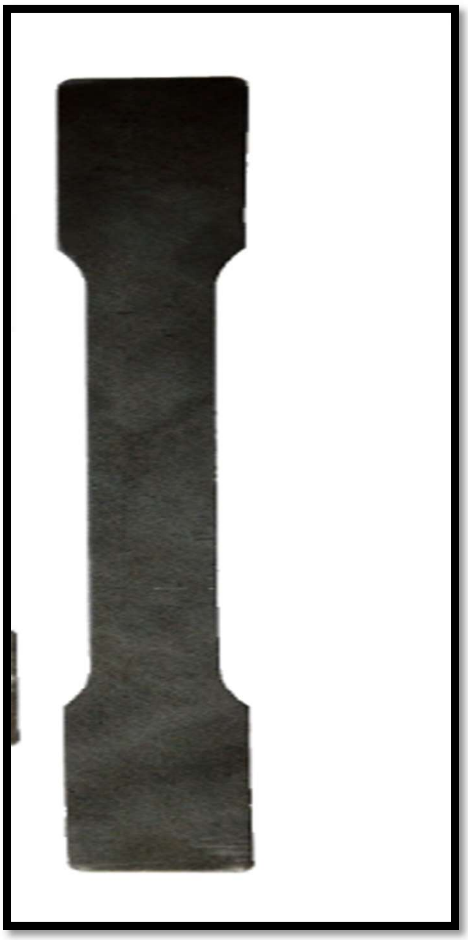

Fig9 conventional sample

3.3 Experimental Tensile Testing

Tensile testing was performed using a universal tensile testing machine under quasi-static loading conditions. The testing setup included a load cell, crosshead displacement system, and digital data acquisition system. The raw experimental output consisted of load data and displacement data, which were converted into engineering stress–strain curves using specimen geometry.

Fig10 ASTM E8

3.4 Engineering and True Stress–Strain Calculations

Engineering stress and strain were calculated using:

$$\sigma = F / A_0 \quad \text{and} \quad \varepsilon = \Delta L / L_0$$

The engineering stress–strain curve showed an initial elastic region, yielding behavior, strain hardening, necking, and final fracture. To improve modeling accuracy during plastic deformation and necking, engineering data were converted into true stress–true strain form:

$$\sigma t = \sigma e(1 + \varepsilon e) \quad \text{and} \quad \varepsilon t = \ln(1 + \varepsilon e)$$

Plastic strain was subsequently calculated by subtracting elastic strain from total true strain:

$$\varepsilon p = \varepsilon t - \sigma t / E$$

These processed true stress–plastic strain values were prepared as constitutive material input for ABAQUS.

3.5 Mechanical Properties Obtained

The processed tensile data provided the mechanical properties shown in Table 2.

Table 2: Experimentally obtained mechanical properties of Zr–2.5%Nb alloy

Property	Value
Yield Stress (0.2% Offset)	618.76 MPa
Yield Stress (Plastic Strain Method)	562.71 MPa
Ultimate Tensile Strength (UTS)	689.10 MPa

Percentage Elongation	15.51%
Reduction in Area	48.36%

3.6 ABAQUS Finite Element Model

The processed true stress–plastic strain data were used for finite element simulation. The preliminary modeling setup included: a conventional dog-bone geometry, elastic-plastic material definition, nonlinear large deformation analysis, and displacement-controlled loading. Hill's anisotropic yield criterion was applied to account for the crystallographic texture of the alloy.

1. Geometry Creation

The tensile specimen geometry was prepared according to standard dimensions used in conventional tensile testing. A dog-bone shaped specimen was modeled in the simulation software.

Objectives

- Create accurate specimen geometry
- Ensure proper meshing region in gauge length
- Prepare model for tensile loading

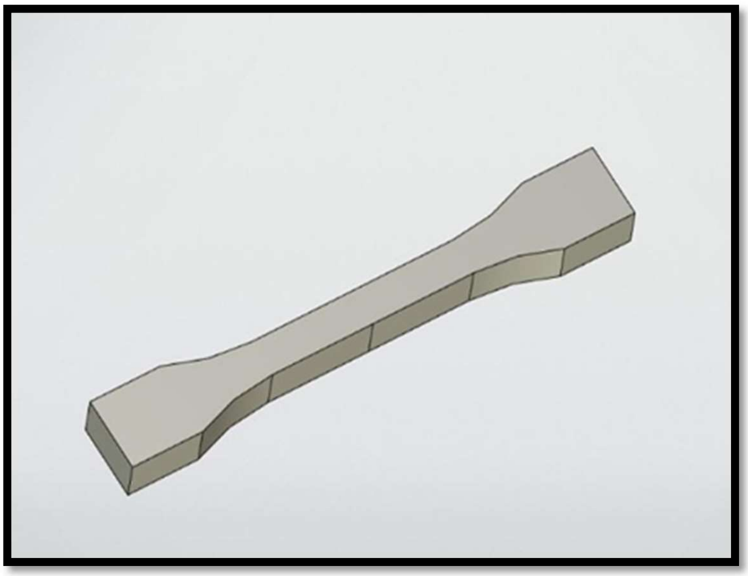

Fig11 modeling sample

2. Material Property Definition

10 The mechanical properties of Zr–2.5Nb alloy were defined using experimental tensile test data. Important properties used in the simulation include:

33 Young's Modulus, Yield Strength, Poisson's Ratio, Plastic deformation data

3. Meshing of Specimen

Meshing divides the specimen into small finite elements for numerical calculations. A finer mesh was applied in the gauge region because maximum deformation occurs there.

Meshing Objectives

- Improve stress accuracy
- Reduce numerical error
- Capture plastic deformation properly

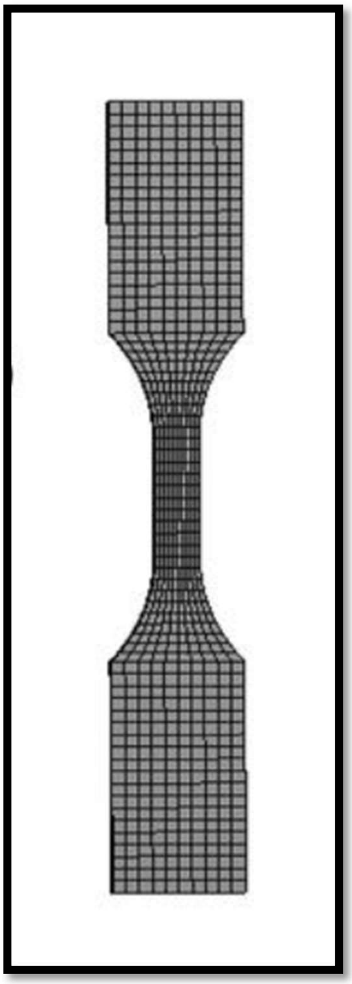

Fig12 mesh image

4. Boundary Conditions and Loading

36 Boundary conditions were applied to simulate the actual tensile test.

Applied Conditions

- One end fixed
- Opposite end subjected to displacement loading
- Tensile load applied along axial direction

The simulation was performed under quasi-static loading conditions.

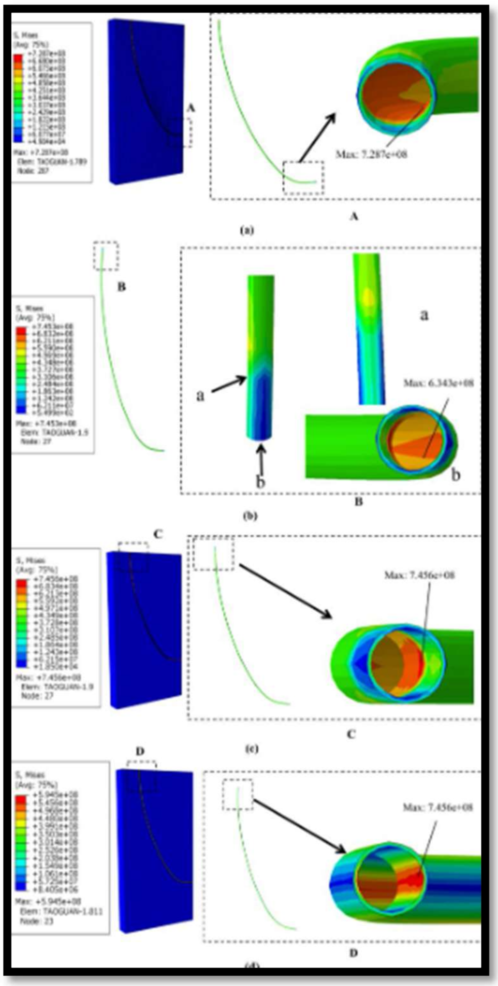

Fig13 boundry condition

5. Stress–Strain Analysis

After simulation, stress–strain curves were obtained and compared with experimental results. The model successfully predicted elastic deformation, yielding behavior, and plastic deformation of the alloy.

Observations

- Stress concentration observed in gauge section
- Uniform deformation before necking
- Good agreement with experimental tensile data

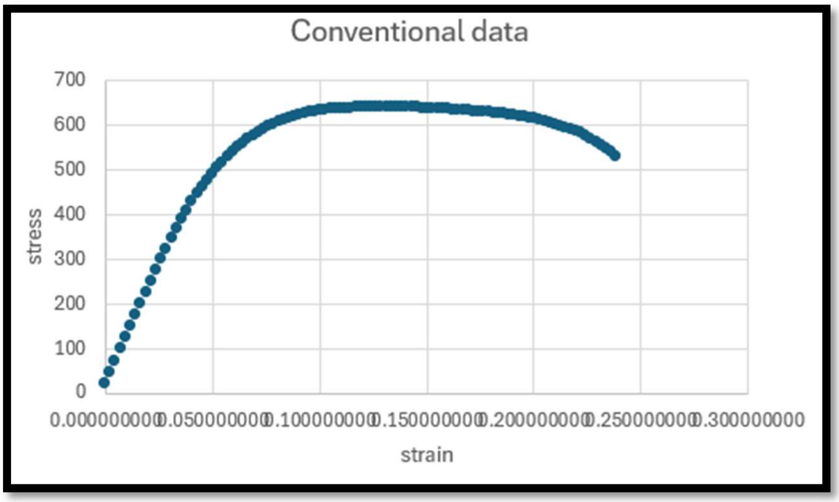

Fig14 stress vs strain

3.7 Comparison of Experimental and ABAQUS Results

Table 3 presents the comparison between experimental tensile properties and ABAQUS simulation predictions.

Table 3: Comparison of experimental and ABAQUS simulation results

Mechanical Property	Experimental Value	ABAQUS Value	Difference (%)
Yield Strength (0.2% Offset)	618.76 MPa	614.25 MPa	0.73%
Ultimate Tensile Strength	689.10 MPa	684.82 MPa	0.62%
Percentage Elongation	15.51%	14.63%	5.67%
Reduction in Area	48.36%	44.91%	7.13%

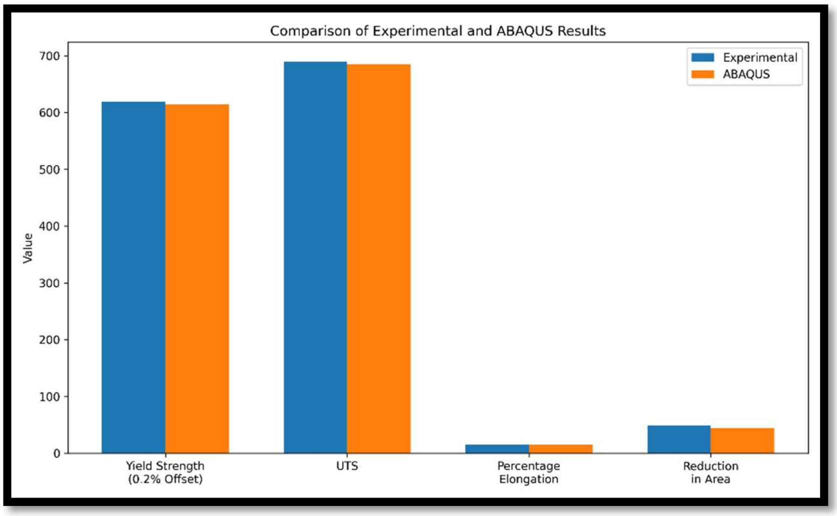

Fig 15compare data

The differences between experimental and simulation results were small: yield strength and UTS deviations were below 1%, while elongation and reduction in area differed slightly more due to necking and fracture complexities. These deviations are attributed to mesh sensitivity, simplified material assumptions, absence of advanced damage models, machine compliance effects, and experimental uncertainties.

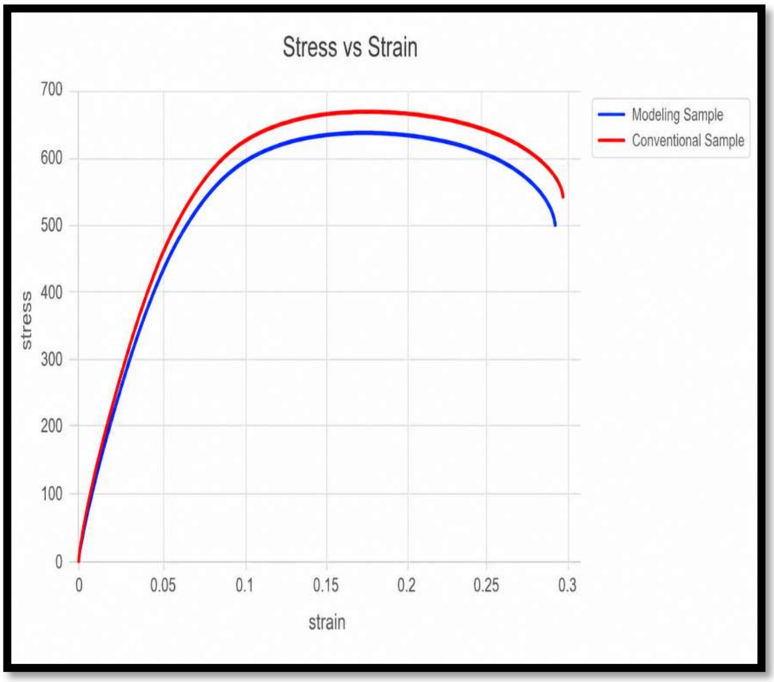

Fig16 Comparision both (stress vs strain)

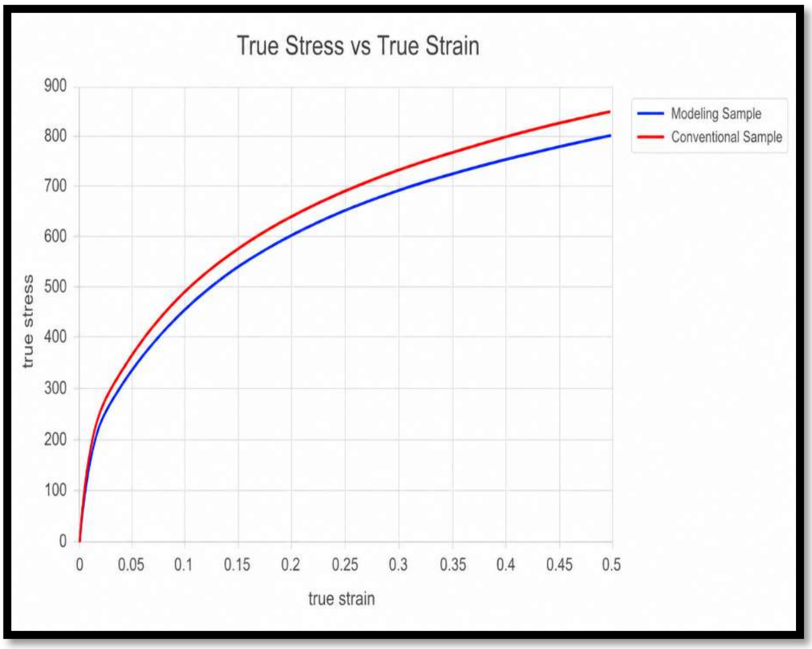

Fig17 comparsion (true stress vs true strain)

3.8 Challenges Encountered

- Accurate yield point determination: The 0.2% offset yield strength required precise elastic slope estimation; small errors in modulus estimation affected the calculated yield strength.
- Necking region accuracy: After necking onset, the stress state becomes nonuniform and engineering stress becomes inaccurate; true stress correction was necessary.
- Plastic strain calculation: Accuracy depended on correct Young's modulus estimation and accurate true strain conversion.
- Experimental noise: Minor fluctuations in experimental data affected logarithmic fitting and strain hardening exponent calculation.

CHAPTER 5

Summary

This progress report focused on the modeling and analysis of conventional tensile testing behavior of Zr–2.5%Nb alloy, which is an important structural material widely used in nuclear reactor pressure tubes due to its excellent corrosion resistance, low neutron absorption cross-section, and good mechanical strength under reactor operating conditions.

The report first discussed the importance of conventional tensile testing as a standardized method for determining key mechanical properties such as yield strength, ultimate tensile strength, ductility, strain hardening behavior, and fracture characteristics. Conventional tensile specimens provide reliable and reproducible data because larger specimen dimensions reduce the influence of microstructural heterogeneity and geometrical inconsistencies. The growing importance of numerical modeling techniques for understanding localized deformation, stress concentration, necking, and fracture mechanisms was also highlighted.

The literature review provided a detailed overview of previous studies related to Zr–2.5%Nb alloy metallurgy, tensile behavior, anisotropy, hydride embrittlement, elevated-temperature performance, and finite element modeling. The alloy was identified as a two-phase α + β zirconium alloy in which niobium stabilizes the β phase and improves irradiation resistance and mechanical strength. Crystallographic texture and phase distribution strongly influence tensile behavior and anisotropy. Temperature significantly affects tensile properties, with strength decreasing and ductility increasing at elevated temperatures due to dynamic recovery and recrystallization.

The preliminary work presented in this report focused on processing experimentally obtained tensile data from conventional tensile testing of Zr–2.5%Nb alloy and preparing the material data for finite element simulation. Standard ASTM tensile specimens (gauge length: 26.37 mm, thickness: 4.57 mm, width: 6.04 mm) were used for testing under quasi-static loading. Experimental load–displacement data were converted into engineering and true stress–strain curves. Mechanical properties extracted from the experimental data were: yield strength ≈ 618.76 MPa, UTS ≈ 689.10 MPa, percentage elongation ≈ 15.51%, and reduction in area ≈ 48.36%. In the true stress–true strain curve, both samples demonstrate continuous strain hardening behavior. The conventional sample remains slightly above the modeling sample, showing higher resistance to deformation at increasing strain levels.

CHAPTER 6

References

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